MIL-C-29363
8 December 1977

MILITARY SPECIFICATION

CLOTH, POPLIN: POLYESTER AND COTTON (WATER REPELLENT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of cloth, poplin, polyester and cotton (water repellent treated).

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-1134	- Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

MILITARY

MIL-T-43548	- Thread, Polyester, Cotton Covered
MIL-T-43624	- Thread, Polyester, Spun

STANDARD

FEDERAL

FED-STD-4	- Glossary of Fabric Imperfections
FED-STD-191	- Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commandant of the Marine Corps (LMO-1), Headquarters, U. S. Marine Corps, Washington, D. C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

FSC 8305

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal shall apply.

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First acticle. When specified (see 6.2), the contractor shall furnish sample cloth for first article inspection and approval. The first article shall consist of 5 yards, full width. The approval of the first article authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all provisions of the specification. The first article shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production cloth.

3.3 Material.

3.3.1 Fibers. The fiber shall be polyethylene glycol terephthalate and cotton. The finished cloth shall contain 65 ±3 percent polyester and the remaining percentage cotton, based on the dry weight of the desized and dyed cloth prior to water repellent treatment, when tested as specified in 4.6.

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3.3.2 Yarns. The warp and filling yarns shall be a blend of 65 percent polyester and 35 percent cotton (see 3.3.1). The cotton portion shall be carded and combed and the blend shall be spun into 2-ply yarn for the warp and a single yarn for the filling. Tests shall be made as specified in 4.6.

3.4 Design and construction.

3.4.1 Physical requirements. The finished cloth shall conform to the requirements shown in Table I when tested as specified in 4.6.

TABLE I. Physical requirements

Weight per sq. yd.		Yarns per inch Minimum		Breaking strength Minimum		Tearing strength Minimum		
Min.	Max.	Warp	Filling	Warp	Filling	Warp	Filling	Weave
Ounces				Pounds		Pounds		
5.5	6.8	110	68	185	70	5.0	2.0	Plain

3.5 Performance.

3.5.1 Dimensional stability. The finished cloth shall not shrink or elongate more than 2.0 percnet in the direction of the warp nor more than 1.0 percnet in the direction of the filling when tested as specified in 4.6.

3.5.2 Water repellency and air permeability. The cloth shall be given an approved flurocarbon type repellent treatment and shall conform to the water repellency and air permeability requirements of Table II and 3.5.3 when tested as specified in 4.6.

TABLE II. Water repellency and air permeability requirements

	Hydrostatic height (centimeters) Minimum	Dynamic absorption (Maximum)	Air permeability (cubic ft. minute/sqft.) Maximum
Initial	35	18	4.0
After 1 dry cleaning	35	--	---
After 3 launderings	32	18	---

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3.5.3 Spray rating. The results of the three individual determinations on the sample unit for spray rating shall be equal to or better than ratings 100, 100, 90 initial and 80, 80, 70 after one dry cleaning when tested as specified in 4.6.

3.5.4 pH. The pH value of the water extract of the finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.6.

3.5.5 Resistance to organic liquid. The finished cloth shall show no wetting by n-tetradecane either initially or after 3 launderings when tested as specified in 4.6.

3.5.6 Seam efficiency. The finished cloth shall have a seam efficiency of no less than 85 percent when tested as specified in 4.6.

3.6 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. The cloth shall be put up on full width rolls as specificed in PPP-P-1134.

3.7 Width. The minimum width shall be 44 inches, inclusive of selvages, unless otherwise specified.

3.8 Color. The color shall be as specified (see 6.2) and shall match the standard sample (see 6.3).

3.8.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.6. When a standard sample is not available, the dyed and finished cloth shall show not more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.6.

3.8.2 Matching. The color of the finished cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.8.3 Colorfastness. The finished cloth shall show fastness to laundering (after 3 cycles), perspiration, crocking wet-dry cleaning, and light equal to or better than the standard sample. When no standard sample is available the dyed and finished cloth shall show "good" fastness to laundering (after 3 cycles), perspiration, crocking, wet-dry cleaning, and light after 40 standard fading hours and shall show a Munsell Value for crocking not lower than 8.5 dry and 7.5 wet when tested as specified in 4.6.

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3.9 Finish. The cloth shall be scoured, heat treated and finished to meet the requirements of this specification. The cloth shall be made water repellent (see 3.5.2). Prior to application of the water repellent treatment, the starch and protein content (including chloroform-soluble and water coluble material) shall not exceed 2.0 percent when tested as specified in 4.6.

3.10 Marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll. Each foll of cloth shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act.

3.11 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When first article is required, it shall be tested as specified in 4.6.

4.5 Quality conformance inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

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4.5.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.5.2 Examination of the end item. Examination of the end item shall be in accordance with 4.5.2.1 through 4.5.2.3.

4.5.2.1 Yard by yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.5.2.1.1 except that only those slubs and knots which exceed the maximum limits shown on Sears Fabric Defect Scales (see 6.5) E for slubs and C for knots shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 25.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds the 38.0 points. If no individual roll exceeds 38.0 points per 100 square yards, the lot shall be acceptable with respect to this characteristic. If one roll exceeds 38.0 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 38.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) X total yards inspected}} = \text{Points per 100 square yards}$$

4.5.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	-	one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

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The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth.
Width less than specified.
Poor dye penetration, mottled, streaky or cloudy.
Overall uncleanness.
Excessive nappiness
Shade defects - shaded side to side, side to center, or end to end.

4.5.2.2 Examination for shade and finish. Each roll in the lot shall be examined visually for shade match (see 3.8). A roll shall be unacceptable if it fails to meet the requirements for shade match. The sample unit shall be a 4-inch by 20-inch swatch of the cloth. The sample unit shall be drawn from each roll in the lot.

4.5.2.3 Examination for length.

4.5.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.5.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.5.2.4 Examination for face marking, and compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Face stamping missing from either of both ends.
Face stamping on wrong side.
Not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.5.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that the packaging, packing and marking complies with Section 5 requirements.

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4.6 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable, and as listed in Table III shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 1/2 yard, full width of the cloth prior to application of the water-repellent treatment for determinations of the nonfibrous materials content and fiber content, and 5 continuous yards full width of the finished cloth for all other physical and chemical tests. The lot size shall be expressed in units of 1 yard each. The lot shall be unacceptable if one or more units fail to meet any test requirements specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.6.1 Fiber content. The general procedure of method 2100 shall be followed and the fiber content percentages shall be calculated as follows:

$$\frac{\text{Weight of dry residual fiber X 100}}{\text{Weight of dry desized specimen}} = \text{percent polyester}$$

100 - percent polyester = percent cotton

Two specimens shall be tested from each sample unit and the average percent cotton of the two specimens shall be reported to the nearest 1.0 percent.

4.6.2 Test for resistance to organic liquid. Place a small specimen of the cloth on a smooth horizontal surface, face up. Using a pipette or eye dropper, gently deposit one drop of n-tetradecane on the surface of the specimen. After one minute, examine the specimen under light at an angle. Absense of light reflectance at the fabric-drop interface shall be taken as evidence of wetting. Three specimens (or areas) taken at various locations across the sample shall be tested. Evidence of wetting on one or more specimens shall be considered cause for rejection of the lot represented by the sample.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

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TABLE III. Test methods

Characteristics	Requirement paragraph	Test Method
Yarn		
Polyester identification	3.3.1	1/
Cotton identification	3.3.1	1200 1/
Fiber content		
Polyester (percent)	3.3.1	2100, 4.6.1 1/
Cotton (percent)	3.3.1	2100, 4.6.1 1/
Yarn		
Warp (2 ply)	3.3.2	Visual 2/
Filling (single)	3.3.2	Visual 2/
Labile sulfur	3.8.1	2020
Colorfastness to:		
Laundering (after 3 cycles)	3.8.3	5610 3/ 4/
Perspiration	3.8.3	5680
Crocking	3.8.3	5651
Wet-dry cleaning	3.8.3	5622
Light	3.8.2	5660
Weight	3.5	5041
Yarns per inch		
Warp	3.4.1	5050
Filling	3.4.1	5050
Breaking strength		
Warp	3.4.1	5100
Filling	3.4.1	5100
Tearing strength		
Warp	3.4.1	5132
Filling	3.4.1	5132
Weave	3.4.1	Visual 2/
Nonfibrous materials	3.6.1	2611 1/
Dimensional stability		
Warp	3.9	5556 5/
Filling	3.9	5556 5/
Hydrostatic pressure		
Initial	3.5.2	5514
After 3 launderings (cotton method)	3.5.2	5556 and 5514 6/
After 1 dry cleaning	3.5.2	5508 and 5514
Dynamic absorption		
Initial	3.5.2	5500
After 3 launderings (cotton method)	3.5.2	5556 and 5500 6/

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TABLE III. Text methods (cont'd)

Characteristics	Requirement paragraph	Test Method
Air permeability		
Initial	3.5.2	5450
Water repellents	3.5.2	7/
Spray rating		
Initial	3.5.3	5526
After 1 dry cleaning	3.5.3	5508 and 5526
pH	3.5.4	2811
Resistance to organic liquid		
Initial	3.5.5	4.6.2
After 3 launderings (cotton method)	3.5.5	5556 and 4.6.2 6/
Seam efficiency	3.5.6	5110 8/

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

3/ The specimen must be dried after each complete laundering cycle.

4/ Only the stain on the polyester and cotton fibers of the transfer cloth shall be evaluated.

5/ Cotton procedure shall be used.

6/ Specimens shall be subjected to 3 complete cycles (wash and dry) prior to determination of hydrostatic height, air permeability, and resistance to organic liquid after laundering.

7/ The contractor shall report the water repellents used, and certify that no other material has been added.

8/ Needle size 0.040 inch (± 0.001) across the blade eye. The thread can be either polyester/cotton in accordance with MIL-T-43548, Ticket No 50, 3 ply for the needle and Ticket No. 70, 2 ply for the looper or spun polyester in accordance with MIL-T-43624, class 1 or 2, Ticket No. 40, 3 ply for the needle and Ticket No. 70, 2 ply for the looper.

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5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for Marine Corps All-weather Coat and lightweight jackets (windbreakers) for men and women.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification.
(b) When first article is required (see 3.2).
(c) Length required if other than specified (see 3.6).
(d) Width of cloth required when other than specified (see 3.7).
(e) Color of cloth required (see 3.8).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.2 Contract data requirements. Data conforming to Data Item Descriptions DI-R-4803, DI-R-4805, DI-T-4901, DI-T-4902, DI-T-4903 and DI-T-4904 will usually be required for delivery in connection with this specification for Marine Corps procurements. For other than Marine Corps, the cognizant procuring activity will provide comparable Data Item Descriptions. When so required, such data will be specified for delivery on a Form DD-1423 included in the contract.

6.3 Standard sample. For access to standard sample (see 3.1), address the procuring office issuing the invitation for bids.

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6.4 Dyestuff formulations. Suggested, but not mandatory, dyestuff formulations are as follows:

Marine Corps Green Shade 2242	Marine Corps Blue Shade 2324
Sodyecron Navy AR	Sodyecron Navy AR
Sodyecron Rubbine EP	Sodyecron Rubbine EP
Orange 62	Bucron Brown Y
Vat Black 25	Disberse Orange 62
Vat Green 3	Sandothrene Blue N2B
Vat Verantherene Brown G	Cibanone Violet BNA
	Vat Violet 13
	Anaphrene Black HBMF

6.5 Defect scales. Sears Fabric Defect Scales are available from Sears, Roebuck and Co., Department 733Q, "Sears Defect Replica Kit", Sears Tower, Chicago, Ill. 60684.

6.6 First article. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

Custodian:
Navy - MC

User Activity:
Army - GL

Preparing activity:
Navy - MC
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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

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6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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